7 points total

1

Chapter 3: 3, 6, 12, 14, 32, 35 (see below for textbook problems)

Additional problems:

1) For the program below, answer the following questions. a) Determine the time delay in ms in the DELAY subroutine. b) What action/operation is performed in the loop AGAIN? Assume that the crystal frequency is 8 MHz and the system is using an ATmega32.

```
.ORG 0
        LDI R16,HIGH(RAMEND)
        OUT SPH,R16
        LDI R16, LOW(RAMEND)
        OUT SPL,R16

        LDI R16,0x40
AGAIN:  COM R16
        OUT PORTD,R16
        CALL DELAY
        RJMP AGAIN
DELAY:  LDI R19,50
LOOP1:  LDI R20,100
LOOP2:  LDI R21,200
LOOP3:  NOP
        DEC R21
        BRNE LOOP3
        DEC R20
        BRNE LOOP2
        DEC R19
        BRNE LOOP1
        RET
```

a) [(1+1+2)200 − 1 + 1 + 2] 100 − 1 + 1 + 2 + 1 + 1 + 2] 50 − 1 + 1 + 4

= ((800+3)100+3)50+4

= 4 015 154 MCS

$t_d = 4\,015\,154\ MC\left(\frac{1\ cc}{1\ MC\ AVR}\right)\left(\frac{1\ s}{8(10^6)\ cc}\right)$ fosc

td = 502 ms (rounded)

b) Complementing 0x40, outputting to PORTD the complemented value, waiting t_d and repeating the process continuously

2) Find the time delay of the following program assuming an oscillator frequency of 4 MHz. Modify the program to achieve the same delay using a nested loop.

2 pts

```
DELAY:  LDI R17, 100
LOOP1:  NOP
        NOP
        NOP
        NOP
        NOP
        NOP
        NOP
        NOP
        NOP
        NOP
        DEC R17
        BRNE LOOP1
        RET
```

LDI Nops DEC BRNE last BRNE RET

1 + (10 + 1 + 2) 100 − 1 + 4

= 1304 MCs

$t_d = 1304\ MCs\left(\frac{1\ cc}{1\ MC\ AVR}\right)\left(\frac{1\ s}{4(10^6)\ cc}\right)$

= 0.000326 sec

t_d = 326 ms

$\mu = 10^{-6}$

Modify the program to achieve the same delay using a nested loop.

1304 MCs

>255
<65025
∴ 1 nested loop

```
1    DELAY:  LDI  R17, Y=25
1    Loop:   LDI  R16, X=16
1    Loop1:  DEC  R16
2/1          BRNE Loop1
1            Nop
1            DEC  R17
2/1          BRNE Loop
1            Nop
4            RET
```

$$1+\left(\left((1+2)x-1\right)+1+1+1+2\right)y-1+\underset{\text{Loop1}}{}+\underset{\text{RET}}{4}+\underset{\text{Nop}}{1}$$

$$(3x+4)y+4=1304$$

$$(3x+4)y=1300$$

Let $y=25$

$$3x+4=52$$

$$x=16$$

Replacing only the 10 NOPs with a loop.
10 MCs = 1 LDI + (1 DEC + 2 BRNE)Q - 1 BRNE
3Q = 10 => Q NOT an integer

```
        LDI, R27,Q
Loop2: DEC R27
        BRNE Loop2
```

So, add a NOP instruction to Loop2.

```
        LDI R27, Q
        NOP
Loop2:  DEC R27
        BRNE R27
```

10 MCs = 1 LDI + 1 NOP + (1 DEC + 2 BRNE)Q - 1 BRNE
3Q = 9 => Q=3
Replace the 10 NOPs with the lines of code above with LDI R27,3

3) Given the following sequence of code. Determine the values of the SP and registers specified after each instruction is executed.

	SP	R16	R24	R25	R26	R27
.ORG 0						
LDI R16, HIGH(RAMEND)	—	08	—	—	—	—
OUT SPH, R16	08	08	—	—	—	—
LDI R16, LOW(RAMEND)	08	5F	—	—	—	—
OUT SPL, R16	085F	5F	—	—	—	—
LDI R24, 0x10	085F	5F	10	—	—	—
LDI R25, 0x48	085F	5F	10	48	—	—
LDI R26, 0x30	085F	5F	10	48	30	—
LDI R27, 0x09	085F	5F	10	48	30	09
PUSH R26	085E	5F	10	48	30	09
PUSH R24	085D	5F	10	48	30	09
PUSH R27	085C	5F	10	48	30	09
POP R24	085D	5F	09	48	30	09
POP R26	085E	5F	09	48	10	09
POP R25	085F	5F	09	30	10	09

Top of RAM = 0x85F

085C	
085D	09
→ 085E	10
085F	30

4) Given the following assembly language instruction sequence and machine codes for the instructions. In the table below, fill in the starting memory location and machine code for each instruction. Show the calculations for the jump instruction to determine the branch addresses as given by the determined machine code values. See Example 3-7.

Starting Code Memory Location for Instruction	Machine Code (see below for different instructions)	Assembly Language Program Instruction
$00000	E450	LDI R21, 0x40
00001	E0F2	LDI R30, 2
00002	0F5E	AGAIN: ADD R21,R30
00003	F7F1	BRNE AGAIN

R21 R30
16 16
15 F14
21 30

1110 0100 0101 0000
1110 0000 1111 0010
0x02

30 → 11110
21 → 10101

ADD R21, R30
0000 11|1 0101, 1110
R21 R30

0F5E

BRNE
1111 01|11 1111 110, 001
F 7 F 1

$PC_{new} = PC_{current} + K + 1$
02 03 $K = -2 \Rightarrow$

00000010
+ 1111 1101
+ 1
11111110

4) Create an Atmel Studio 7 workspace for the ATmega32. Copy the code from Additional Problem 3 above. Single Step execute the program and follow the values of the General Purposes Registers (R16, R24-R27) (you can either do this by showing the GPRs or by tracking the SRAM locations 0x10, 0x18-0x1B). Verify that your answers in the table above are correct. Submit a screen shot of your program simulation after executing the final instruction POP R25.

Screen shot submission of Memory (Data) from 0x00-0x1B or screen shot submission of workspace with Processor Status showing GPR values and SP.

6)

5)

```
        .ORG 0
        LDI  R30, 0       ; low byte
        LDI  R31, 0       ; high byte initialization
        LDI  R16, 0x3C    ;   add 1st byte to low byte
        ADD  R30, R16     ;   sum                      (number)
        BRSH Nxt_Number   ; if C=0, go to next number
        INC  R31          ; C=1 => inc high byte of sum
Nxt_Number: LDI R16, 0x57 ;
        ADD  R30, R16     ; add 2nd number to low byte
                          ; sum
        BRSH Nxt_Number2  ; if C=0, go to next
                          ; number
        INC  R31          ; C=1 => inc high byte
                          ; of sum
Nxt_Number2: LDI R16, 0xA6
        ADD  R30, R16     ; add 3rd number to low
                          ; byte sum
        BRSH Done         ; if C=0, done adding
                          ; numbers
        INC  R31          ; C=1, update high byte
                          ; of sum
Done:   JMP  Done         ; terminal program
                          ; statement
```

BRSH not required heresince 0+8 bit number cannot be > 8

STS k, Rr ;Store register Rr in memory location k

1001	001r	rrrr	0000
kkkk	kkkk	kkkk	kkkk

$0 \le r \le 31, 0 \le k \le 65535$

ADD Rd, Rr ;Add Rd to Rr

0000	11rd	dddd	rrrr

$0 \le d \le 31, 0 \le r \le 31$

LDS Rd, k ;Load from memory location k to register Rd

1001	000d	dddd	0000
kkkk	kkkk	kkkk	kkkk

$0 \le d \le 31, 0 \le k \le 65535$

LDI Rd, K ;load register Rd with value K

1110	KKKK	dddd	KKKK

$16 \le d \le 31, 0 \le K \le 255$

BRNE BranchPlace; conditional branch -> branch if C = 0

1 1 1 1	0 1 K6 K5	K4 K3 K2 K1	K0 0 0 1

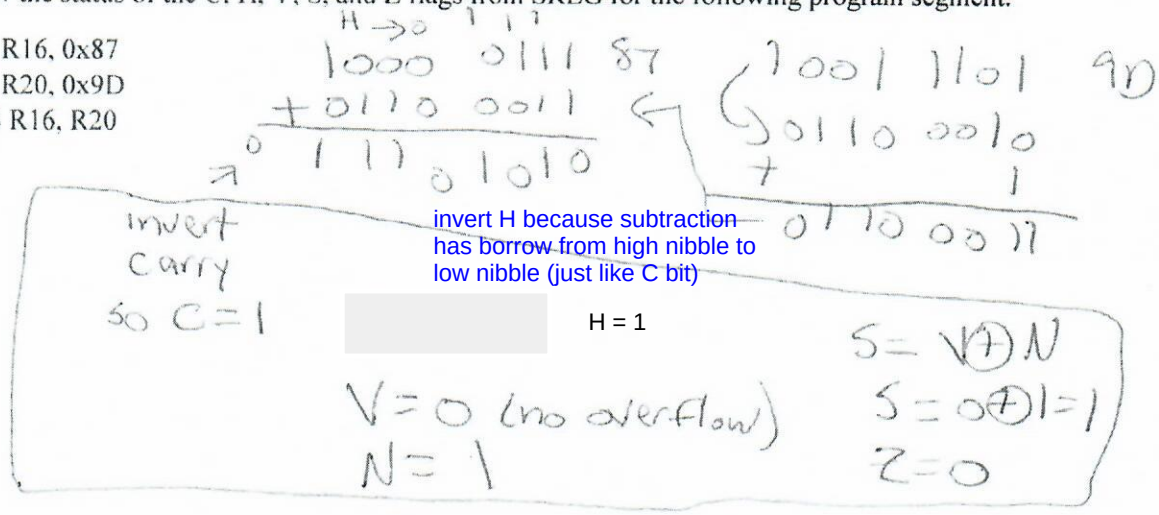
K6 K5 K4 K3 K2 K1 K0 = > -64 to +63 (2s complement range) offset with respect to the current PC

6) 5) Write an assembly language program for the ATmega32 to compute the sum of 0x3C, 0x57, 0xA6. Store the result in registers R30 (low byte) and R31 (high byte). See example 3-5 previous

See next page

7) 6) Show the status of the C, H, V, S, and Z flags from SREG for the following program segment.

LDI R16, 0x87
LDI R20, 0x9D
SUB R16, R20



17. CALL is a(n) ___-byte instruction.
18. RCALL is a(n) ___-byte instruction.
19. True or false. The RCALL target address can be anywhere in the 4M (word) address space.
20. True or false. The CALL target address can be anywhere in the 4M address space.
21. When CALL is executed, how many locations of the stack are used?
22. When RCALL is executed, how many locations of the stack are used?
23. Upon reset, the SP points to location _____.
24. Describe the action associated with the RET instruction.
25. Give the size of the stack in AVR.
26. In AVR, which address is pushed into the stack when a call instruction is executed.

SECTION 3.3: AVR TIME DELAY AND INSTRUCTION PIPELINE

27. Find the oscillator frequency if the machine cycle = 1.25 μs.
28. Find the machine cycle if the crystal frequency is 20 MHz.
29. Find the machine cycle if the crystal frequency is 10 MHz.
30. Find the machine cycle if the crystal frequency is 16 MHz.
31. True or false. The CALL and RCALL instructions take the same amount of time to execute even though one is a 4-byte instruction and the other is a 2-byte instruction.
32. Find the time delay for the delay subroutine shown below if the system has an AVR with a frequency of 8 MHz:

```
1          LDI     R16, 200
1  BACK:   LDI     R18, 100
1  HERE:   NOP
1          DEC     R18
2/1        BRNE    HERE
1          DEC     R16
2/1        BRNE    BACK
```

(1+1+2)100 − 1 = X (BRNE last)

(1+X+1+2)200 − 1 (BRNE last) + LDI 1st instruction

X = 399

(1+399+1+2)200 − 1 + 1 = 80600 MCs

$80600\,MCs\left(\frac{1\,cc}{1\,MC}\right)\left(\frac{1\,s}{8(10^6)\,cc}\right) = 0.01\,sec$

SECTION 3.1: BRANCH INSTRUCTIONS AND LOOPING

1. In the AVR, looping action with the "BRNE target" instruction is limited to ____ iterations.
2. If a conditional branch is not taken, what is the next instruction to be executed?
3. In calculating the target address for a branch, a displacement is added to the contents of register PC.
4. The mnemonic RJMP stands for _____ and it is a(n) ___-byte instruction.
5. The JMP instruction is a(n) ___-byte instruction.
6. What is the advantage of using RJMP over JMP? Uses less code space than JMP
7. True or false. The target of a BRNE can be anywhere in the 4M word address space.
8. True or false. All AVR branch instructions can branch to anywhere in the 4M word address space.
9. Which of the following instructions is (are) 2-byte instructions.
   (a) BREQ (b) BRSH (c) JMP (d) RJMP
10. Dissect the RJMP instruction, indicating how many bits are used for the operand and the opcode, and indicate how far it can branch.
11. True or false. All conditional branches are 2-byte instructions.
12. Show code for a nested loop to perform an action 1,000 times.
13. Show code for a nested loop to perform an action 100,000 times.
14. Find the number of times the following loop is performed:

```
        LDI     R20, 200
BACK:   LDT     R21, 100
HERE:   DEC     R21
        BRNE    HERE
        DEC     R20
        BRNE    BACK
```

RJMP 2 bytes
JMP 4 bytes

200x100 = 20000 times

12)

```
        LDI R17,100
Loop2:  LDI R16,100
Loop1:  DEC R16
        BRNE Loop1
        DEC R17
        BRNE Loop2
```

1000 = 100x100
≤255
many possibilities

35. Find the time delay for the delay subroutine shown below if the system has an AVR with a frequency of 10 MHz:

```
1          LDI     R20, 200
1  BACK:   LDI     R25, 100
1          NOP
1          NOP
1          NOP
1  HERE    DEC     R25
2/1        BRNE    HERE
1          DEC     R20
2/1        BRNE    BACK
```

DEC BRNE (1+2) 100 − 1 (last BRNE) = 299 = X

LDI Nop Nop Nop X DEC BRNE (last BRNE)

$(1+1+1+1+299+1+2)200-1$

$= 61200$

$t_d = 61200\ \text{MCs}\left(\frac{1\text{cc}}{1\text{MC}}\right)\left(\frac{1\text{s}}{10^6\text{cc}}\right) = 0.00612$

$6.12\ \text{ms}$